## 6.5 Trial Running

Following satisfactory completion of the System Acceptance Tests and the Integrated System Tests the Employer will commence an extended period of Trial Running to prove all technical systems in timetable operation, to allow all technical systems to settle and to train staff in working procedures.

Trial Running shall include:

- The regular circulation of a full complement of trains as required for 24 hours of scheduled service, including peak demand. Intentional disruption of the service shall be included (e.g. extended dwell times, vehicle, power systems, ATP, ATO, point operation and train detection failures) in order to check operational stability and safety of the system and the effectiveness of the supervisory control software employed in reducing the effect of such disruptions;
- The determination of the actual headway achieved at each station for all specified routes and including intermediate reversing movements and movement into and out of the depot.

The Contractor shall allow for attendance over the whole of this period, which may be expected to include maintenance and repair activities and also further opportunity for technical staff training.

## 6.6 Records of Tests

Records of Tests carried out shall be kept by the Contractor and a report and all Test results shall be submitted to the Engineer no later than 30 days after completion of the Test. In addition to any other requirements, the report shall contain the following details:

- material or part of the Works tested;
- location of the part of the Works;
- place of testing;
- date and time of tests;
- technical personnel supervising or carrying out the tests;
- equipment used and method of testing;
- readings and measurements taken during the tests;
- test results, including any calculations and graphs;
- other details stated in the Contract.

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## 6.7 OPERATION & MAINTENANCE

**General**

The Contractor shall ensure that the design of the software and hardware of the Environmental Control System is supportable throughout the service life of the System to address, as a minimum, the following:

design errors in the System;

operational changes;

environmental changes; and

changes in infrastructure.

**During the Defects Liability Period all maintenance (including break down) will be conducted by the contractor and operation will be by the employer staff. Contractor has to train the employer's staff during the course of maintenance.**

The Contractor shall ensure that sufficient no. of personnel are available with the relevant skills and level of competence.during the DLP and maintenance period .

During the DLP period contractor has to provide free of cost all material including consumables.

### 6.7.2 Maintenance During Defects Liability Period

1 Competency of Personnel

During the DLP the Contractor shall support the Authority with sufficient trained and competent personnel.

Such persons shall have their generic competence established and must demonstrate their specific competence and knowledge in the particular systems, environment and procedures.

The Contractor shall provide evidence of specific competence and knowledge, which shall include:

assessment and certified training in particular software applications and operations;

recording of competence and work in the license holders logbook; and

receiving or in receipt of sufficient and current exposure to the area of work that the holder is licensed for.

Routine spot checks on licensing may be carried out from time to time by the Employer's Representative's qualified personnel on the proficiency of the Contractor's staff.

2 Defects Liability Management Plan

As part of the Defects Liability Management Plan, the Contractor shall detail the management and organisation to be provided during the DLP.

3 Discrepancies between Installation and Design Records

Should the Contractor discover inconsistencies between the maintenance drawings and documentation and the installed equipment, the Contractor shall correct all such errors within two weeks.

4 Penalty Clause

Penalty of Rs. 10,000 per day in Maintenance Period will be imposed if major equipment and systems are not working for more then 24 Hrs.

**6.7.2.1 Support Documentation**

**Routine and Corrective Maintenance Procedures**

Routine and corrective maintenance procedures shall be supplied for all equipment. The format shall be as follows:

uniform format and layout irrespective of equipment supplier;

colour coding for each activity;

cross referenced to the Operation and Maintenance Manuals; and

document control information.

The procedures shall be submitted for review by the Employer's Representative; the following shall be included as a minimum:

frequency of maintenance;

type of maintenance;

the equipment identification;

safety precautions to be observed;

step by step guide to the maintenance required; and

explanatory diagrams. Also in each plant room maintenance schedule shall be displayed.

## 6.8 SPARE PARTS, SPECIAL TOOLS & TEST EQUIPMENT : GENERAL

During the Defects Liability Period, the Contractor shall provide free of cost all materials including consumables, unit exchange spares and emergency spares required for maintenance (routine and breakdown) of the Electrical and Mechanical Systems (EMS). EMS also include Station Environmental Control and Building Management System provided under the Contract. The provision of spares during the DLP excludes any spares / consumable required for routine / breakdown maintenance of operation. Before the commission of revenue services the contractor shall supply the spares, jigs and fixtures materials not later than 6 weeks before the commissioning of revenue services. A list of such spares and materials required for maintenance during the Defects Liability Period and to be provided free of cost by the Contractor and if list is found insufficient then the same need to be augmemented for preventive and corrective maintenance of all the equipments supplied under the contract. Without any extra cost.

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If these spares are not consumed during the Defects Liability Period, these shall become the property of the Employer at the end of Defects Liability Period.

### 6.8.1 Tools and Test Equipment

The Contractor shall bring free of cost six weeks before start of trial running special tools and test equipment which are essential for day to day use in both corrective and preventative maintenance and for workshop use in the overhaul of all modules and units likely to be required over the full service life of the installation. The Contractor shall submit a schedule of all tools and equipment with details of calibration and supplier along with the tender. If list is found insufficient then the same need to be augmemented for preventive and corrective maintenance of all the equipments supplied under the contract.

If these tools and equipments shall become the property of the Employer at the end of Defects Liability Period.

### 6.8.2 Spares List

The Contractor shall submit during the period of contract a schedule of spare parts required for EMS duly indicating, for each item of spares, its description, part number, drawing number, lead time, shelf life and number of units required for the ten years (beyond DLP) as well as for the expected life of EMS, principal as well as secondary sources of supply and also the unit price of each spare with escalation clause if list is found insufficient then the same need to be augmemented for all the equipments supplied under the contract.

This schedule shall include all types of consumable, unit exchange and emergency spares. The Contractor shall also advise upon recommended inventory having regard to the lead time of the respective items.

The Employer shall, during a period of ten years from the date of taking over of the whole works, purchase as many parts as required by him, at the rates indicated in this schedule.

If during the period of ten years, the Contractor intends to discontinue the manufacture of spare or replacement parts for the EMS the Contractor shall immediately give notice to the Employer of such intention. The Employer shall be given the opportunity of ordering at reasonable prices such quantities of such spare or replacement parts as the Employer shall reasonably require in relation to the anticipated life of the EMS.

In the event of Contractor failing to supply the spare parts in accordance with this Clause, he shall in respect of each item of spare, furnish free of cost to the Employer, the drawings, specifications, patterns and other information to enable the Employer to make or have made such spare parts. The Employer shall be entitled to retain the aforesaid drawings etc., for such time only as is necessary for the exercise by the Employer of his rights under this clause and the drawings, if the Contractor so requires, shall be returned by the Employer to the Contractor in good order and condition (fair wear and tear excepted).

Under such circumstances, the Contractor shall also grant to the Employer, without payment of any royalty or charge, full right and liberty to make or have made spare or replacement parts as aforesaid and for such purposes only to use, make and have made copies of all drawings, patterns, specifications and other information supplied by the Contractor to the Employer pursuant to the Contract.

The Contractor will so far as it is reasonably able to bind his sub-contractors to conform with the requirements of this Clause and shall, prior to entry into any sub-contracts, provide the Employer with full details of any sub-contractor who will not so conform in which event the Employer may direct the Contractor to seek an alternative sub-contractor.

If the Contractor fails to provide spare or replacement parts as described in this Sub-clause and these are available from the Contractor's sub-contractor, the Employer shall have the right to obtain such spare and replacement parts from the sub-contractor or any other supplier and any additional cost incurred by the Employer shall be recoverable from the Contractor.

The Employer may require the Contractor to enter into a Maintenance Contract with the Employer for the EMS provided under the Contract under terms and conditions to be mutually agreed.

If due to up gradation or advance in technology any new type of models versions or design of spare parts are developed in future, the same shall be plug – compatible and space-compatible with regard to original design and installation of EMS.

Where the Contractor considers that any equipment that would be supplied, and which he considers cannot be economically or technically maintained by the Employer (e.g. computer processors) then such items shall be identified and proposals made for the maintenance of such equipment through OEM's.

All spare parts as mentioned in Clause 6 above shall be provided by the Contractor 6 weeks before commencement of trial running.

The Contractor shall:

submit to the Employer's Representative a list of spares required for the life of the System.

base the spares calculations on the reliability and availability data and the criticality of the equipment.

submit to the Employer's Representative for review the calculations and spares list.

Submit to the Employer's Representative a Cardex system for easy identification of spares.

The Spares list shall:

be grouped [illegible] system, test equipment and special tools as applicable for stocking identification.

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have detailed description with drawing references and correlation with the maintenance manuals.

**6.8.3 Second Sourcing**

The Contractor shall identify principal and second-source suppliers that can supply the systems and sub-system spares listed. The Contractor shall make the second-source supplier information available to the Employer's Representative at the time of submission of the final design.

**6.8.4 Long Lead Times**

The Contractor shall identify the lead times for all spare parts. Parts with long lead times shall be identified as such to the Employer's Representative in the spares list.

**6.8.5 Routine Change**

In the event that any item of the supply requires to be routinely changed or calibrated, regardless of whether it appears in the spares list or not, it shall be identified to the Employer's Representative together with the routine change interval.

**6.8.6 Shelf Life**

In the event that any of the spares identified have a particular shelf life or special storage requirement, this shall be made known to the Employer's Representative with the submission of the spares list, including the necessary action for disposal or storage.

**6.8.7 Identification and Configuration Control**

All spare equipment identified on the spares list, shall conform to Identification and Configuration Control requirements established by the Contractor for the equipment provided under the Contract..

**6.8.8 Testing of Spares**

The Contractor shall ensure that all spares are correctly calibrated, tested and labelled prior to their delivery. Test certificates for each equipment shall be submitted to the Employer's Representative.

**6.9 - TRAINING**

6.9.1 General

Training shall be provided for Employer's engineers and staff such that the Systems can be operated and maintained in accordance with the Manufacturer's requirements.

The Contractor shall submit for review by the Employer's Representative a training plan 180 days prior to the first testing and commissioning activity.

6.9.2 Scope Of Training

The training shall include both normal and abnormal operations.

1. **Operations staff will include as a minimum:-**



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train operators;
traffic controller;
duty chief controller;
depot and station controllers;
Engineering co-ordinator;
communications co-ordinator;
operations planners and scheduler; and
Employer's Training Instructors and trainers.

2 **Engineering staff shall include as a minimum:-**

maintenance staff;
fault report centre staff;
design staff; and
Employer's Training Instructors and trainers.

3 **Management staff shall include as a minimum:-**

supervisors;
line managers;
section managers; and
department managers.

4 **Normal and Abnormal Operation**

The Contractor shall train the Employer's engineers and staff in the normal day to day operation of the Environment Control System.

The Operations and Engineering staff shall also be trained to address efficiently and effectively failures of equipment, software and operation of the Tunnel Ventilation System in these abnormal conditions. The Contractor shall, as a minimum, provide the following:

recovery to normal operations after a System failure and other incidents;
selection and deselecting of backup control functions;
data logging, retrieval and preservation;
contingency arrangements; and
training scenarios for all operational aspects.

The Contractor shall conduct at least 1 course on the operations aspects of the Tunnel Ventilation System. The courses will be attended by a minimum of 10 Employer's staff.

5 **System Design and Configuration**

The Contractor shall provide training to the Employer's engineers and staff in the design and configuration of the System. This training shall include:

Systems overview;

systems configuration and data preparation; and

principles of operation.

6 **Maintenance Training**

Training shall be provided to the Employer's engineers and staff to undertake maintenance of the System provided following substantial completion of the Works. This training shall include:

(1) principles of operation;

(2) preventative and corrective maintenance tasks and procedures;

(3) fault repair to the lowest level replaceable unit;

(4) use of test equipment, diagnostic and maintenance aids; and

(5) software maintenance including database structure, generation, modification and system software organisation.

The Contractor shall conduct at least 2 courses on the maintenance aspects of the Tunnel Ventilation System. The courses will be attended by about 20 Employer's staff.

### 6.9.3 Training Format

The Contractor shall comply with relevant clause of Employer's Requirements - Manufacturing, Installation and Testing.

## 6.10 - DOCUMENTATION

6.10.1 General Requirements

This section defines how submissions, including Design Documents and other material submitted for review by the Employer's Representative, shall be presented and controlled.

6.10.2 Submission Control

The Contractor shall submit a Submissions Programme in accordance with the Employer's Requirements. In addition, the Submissions Programme shall:

(1) identify all design, manufacturing, testing, operations and maintenance contract deliverables, required by the Specification; and

(2) assign reference numbers to all submissions.

All documentation shall be submitted to the format contained in the Employer's Requirements.

Textual submittals, including reports, specifications, and calculations, shall also be submitted in electronic form, wherever feasible.

All 'As built' final documentation, including drawings and support documentation, shall be electronically stored on CD ROM media. The Contractor shall, not less than 90 days prior to the substantial completion of the last Section of each Corridor, submit a first draft of the documentation CD to the Employer's Representative. A final version shall be submitted at the time when the Contractor applies to the Employer's Representative for a Substantial Completion Certificate. Any updates due to design changes during testing or revenue service shall be at the Contractor's expense.

The Standard CD ROM format shall be as defined in ISO 9660. The Contractor shall demonstrate that drawings reproduced from the CD-ROM are legible. Legibility will be determined by being able to distinguish all dimensions, dimension lines, outlines and dashlines without use of supplementary viewing aids.

The CD ROM format shall be consistent with the CAD system specified by the Authority at the time of delivery.

### 6.10.3 Contractor's Responsibilities

In addition to the requirements of the Employer's Requirements, the Contractor shall ensure that the requirements contained herein are carried out.

The Contractor shall maintain a complete up-to-date, organised file of all past and current submissions including an index and locating system, which identifies the status of each submission.

The index shall be available for the Employer's Representative's review, and shall be used to:

(1) assign sequential numbers to each contract deliverable; and

(2) assign new deliverable numbers to all resubmissions and cross-reference to previous submissions.

The Contractor shall provide supplemental information with each submission, in sufficient detail to explain completely the equipment described and their intended manner of use.

### 6.10.4 Material Contract Deliverables

Materials deliverables shall be identified by a clear and durable identification plate, which shall include the following information:

(1) issue and revision status and date;

(2) Contract title and number;

(3) the names of the Contractor, subcontractor, supplier or manufacturer;

(4) identification of product by either description, model number, type number or serial number; and

(5) subject identification by Contract Drawing, Design Drawing or Specification reference.

### 6.10.5 Supplemental Modification Deliverables

Changes in submissions previously reviewed without objection will not be permitted unless those changes are resubmitted in the same manner as the original deliverable and reviewed by the Employer's Representative.

### 6.11 Maintenance beyond Defects Liability Period

DMRC reserves the right to award the annual maintenance contract beyond the defect liability period for 3 years through a supplementary agreement.